#In this code we will demonstrate the lema 7.15. We will check that the component ñj is positive along the γ3 *curve* and *along the* γ4 *curve the component* ñj change the sing exactly once.

#The condition ñj = 0 implies x = t, *this way the constraint equation* (7.1) *is rewritten with*

#Define vj

$vj := 4 \cdot x^2 - (4 \cdot z^2 + 2 \cdot y \cdot x - y^2);$

$$vj := 4x^2 - 2yx + y^2 - 4z^2 \quad \textbf{(1)}$$

#We start with γ3

#Apply the rational parametrization constructed in this chapter 4..

$$parametrization := \left\{ \mathrm{w} = \mathrm{a} + \frac{1}{a}, \right.$$
$$z = b + \frac{1}{b},$$
$$y = a - \frac{1}{a} + b - \frac{1}{b},$$
$$\left. x = a - \frac{1}{a} - b + \frac{1}{b} \right\}:$$

$vj := simplify(numer(\mathrm{expand}(\mathrm{subs}(\mathrm{parametrization, vj}))));$

$$vj := 3a^4b^2 + (-6b^3 + 6b)a^3 + (3b^4 - 28b^2 + 3)a^2 + (6b^3 - 6b)a + 3b^2 \quad \textbf{(2)}$$

#Define r3

$$\begin{aligned} r3 := \big(& a^{22}b^{10} + (-6b^{11} + 8b^9)a^{21} + (15b^{12} + 6b^{10} + 30b^8)a^{20} + (-18b^{13} - 102b^{11} + 162b^9 \\ &+ 74b^7)a^{19} + (3b^{14} + 198b^{12} + 93b^{10} + 600b^8 + 145b^6)a^{18} + (24b^{15} - 96b^{13} - 762b^{11} \\ &+ 1884b^9 + 1380b^7 + 252b^5)a^{17} + (-294b^{16} - 1740b^{14} - 3162b^{12} - 2920b^{10} + 1908b^8 \\ &+ 1092b^6 + 145b^4)a^{16} + (24b^{17} - 132b^{15} - 2472b^{13} - 8010b^{11} + 3340b^9 + 4692b^7 \\ &+ 1380b^5 + 74b^3)a^{15} + 3b^2(b^{16} - 580b^{14} - 3862b^{12} - 8784b^{10} - 8307b^8 - 1172b^6 \\ &+ 636b^4 + 200b^2 + 10)a^{14} + (-18b^{19} - 96b^{17} - 2472b^{15} - 14520b^{13} - 34416b^{11} \\ &- 8268b^9 + 3340b^7 + 1884b^5 + 162b^3 + 8b)a^{13} + (15b^{20} + 198b^{18} - 3162b^{16} - 26352b^{14} \\ &- 67416b^{12} - 68232b^{10} - 24921b^8 - 2920b^6 + 93b^4 + 6b^2 + 1)a^{12} - 6b(b^{20} + 17b^{18} \\ &+ 127b^{16} + 1335b^{14} + 5736b^{12} + 11692b^{10} + 5736b^8 + 1335b^6 + 127b^4 + 17b^2 + 1)a^{11} \\ &+ b^2(b^{20} + 6b^{18} + 93b^{16} - 2920b^{14} - 24921b^{12} - 68232b^{10} - 67416b^8 - 26352b^6 \\ &- 3162b^4 + 198b^2 + 15)a^{10} + (8b^{21} + 162b^{19} + 1884b^{17} + 3340b^{15} - 8268b^{13} - 34416b^{11} \\ &- 14520b^9 - 2472b^7 - 96b^5 - 18b^3)a^9 + (30b^{20} + 600b^{18} + 1908b^{16} - 3516b^{14} \\ &- 24921b^{12} - 26352b^{10} - 11586b^8 - 1740b^6 + 3b^4)a^8 + (74b^{19} + 1380b^{17} + 4692b^{15} \\ &+ 3340b^{13} - 8010b^{11} - 2472b^9 - 132b^7 + 24b^5)a^7 + (145b^{18} + 1092b^{16} + 1908b^{14} \\ &- 2920b^{12} - 3162b^{10} - 1740b^8 - 294b^6)a^6 + (252b^{17} + 1380b^{15} + 1884b^{13} - 762b^{11} \\ &- 96b^9 + 24b^7)a^5 + (145b^{16} + 600b^{14} + 93b^{12} + 198b^{10} + 3b^8)a^4 + (74b^{15} + 162b^{13} \\ &- 102b^{11} - 18b^9)a^3 + (30b^{14} + 6b^{12} + 15b^{10})a^2 + (8b^{13} - 6b^{11})a + b^{12}\big): \end{aligned}$$

#Calculate the resultant of r3 and vj with respect to b

$$R := resultant(r3, vj, b) :$$

#We will chack that R does not have roots bigger than 2.75.

#We apply the translation a=2.75+c as a change of variable

$$translation := \left\{ a = \frac{11}{4} + c \right\} :$$

$$\mathrm{R\grave{}} := \mathrm{collect}(\mathrm{numer}(\mathrm{expand}(\mathrm{subs}(\mathrm{translation}, \mathrm{R}))), c) :$$

#Compute the degree of R\`in c

$$degree(R\grave{}, c);$$

$$132 \tag{3}$$

$$seq(sign(coeff(R\grave{}, c, i)), i = 132..0, -1)$$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, \\ 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, \\ 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, \\ 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \tag{4}$$

#This prove that the possibles intersection by r3=0 with vj=0 occur in the retangle [1 , 2.75) $\times [1, 2.75)$

#To discard completaly the intersections, we apply the moebius transformations to this retangle in r3

$$moebius := \left\{ a = \frac{1 + \frac{11}{4} \cdot c}{1 + c}, b = \frac{1 + \frac{11}{4} \cdot d}{1 + d} \right\} :$$

$$\mathrm{r\grave{3}} := \mathrm{collect}(\mathrm{numer}(\mathrm{expand}(\mathrm{subs}(\mathrm{moebius}, \mathrm{r3}))), c) :$$

#Compute the degree of r\`3 in c

$$degree(r\grave{3}, c);$$

$$22 \tag{5}$$

#Define the coefficients ci = $coeff(r\grave{3}, c, i)$

```
for i from 0 to 22 do
  c[i] := coeff(`r3`, c, i);
od:
```

#Compute the all degree of ci in d

```
seq(degree(c[i], d), i = 0 ..22);
```

22, 22, 22, 22, 22, 22, 22, 22, 22, 22, 22, 22, 22, 22, 22, 22, 22, 22, 22, 22, 22, 22, 22 **(6)**

#From here we will analyze the signal variation of all coefficients ci

```
seq(seq(sign(evalf(coeff(c[j], d, i))), i = 22 ..0, -1), j = 22 ..0, -1);
```

−1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, **(7)**
−1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1,
−1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1,
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−1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1,

$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1$

#This prove that nj have the constant sing along the` $\gamma 3$.

#To finish, we calculate the nj $(\gamma 3(1), 1)$ *sing*

#Define nj

$nj := w^{-3} + z^{-3} - 2 \cdot x^{-3}$

$$nj := \frac{1}{w^3} + \frac{1}{z^3} - \frac{2}{x^3} \tag{8}$$

ñj := *simplify*(*numer*(expand(subs(parametrization, *nj*)))) ;

$$nj := a^{12} b^6 + 3 (-b^7 + b^5) a^{11} + 3 (b^8 - 2 b^6 + b^4) a^{10} + (-2 b^9 - 3 b^7 - 3 b^5) a^9 + 3 (-b^{10} - 5 b^6 + 4 b^4 + b^2) a^8 + 3 (b^{11} - 3 b^9 - 6 b^7 - 16 b^5 - b^3 + b) a^7 + (-b^{12} + 6 b^{10} + 15 b^8 - 15 b^4 - 6 b^2 + 1) a^6 + 3 (-b^{11} - 3 b^9 + 4 b^7 - 6 b^5 - b^3 - b) a^5 + 3 (-b^{10} - 4 b^8 + 5 b^6 + b^2) a^4 + (-4 b^9 - 9 b^7 - 9 b^5 - 2 b^3) a^3 + 3 (-b^8 + 2 b^6 - b^4) a^2 + 3 (-b^7 + b^5) a - b^6 \tag{9}$$

#Check that 2.75 $< \gamma 3 (1) < 3$

$sturm\left(sturmseq(eval(r3, b=1), a), a, \frac{11}{4}, 3\right)$; #number of roots in (2.75 , 3)

$$1 \tag{10}$$

#Chech that ñj $(a, 1)$ *have constant sing* for $2.75 < a < 3$

$sturm\left(sturmseq(eval(\tilde{n}j, b=1), a), a, \frac{11}{4}, 3\right)$; #number of roots in (2.75 , 3)

$$0 \tag{11}$$

#Find the ñj $(\gamma 3(1), 1)$ *s ing through* $nj \cdot (3, 1)$

$eval(eval(\tilde{n}j, b=1), a=3)$

$$190592 \tag{12}$$

#This prove that the component ñj is positive along the $\gamma 3$

restart;

#For γ4, we will apply the parametrization in vj

$vj := 4 \cdot x^2 - (4 \cdot z^2 + 2 \cdot y \cdot x - y^2)$:

$parametrization := \left\{ \mathrm{w} = \mathrm{a} + \frac{1}{a}, \right.$
$z = b + \frac{1}{b},$
$x = a - \frac{1}{a} + b - \frac{1}{b},$
$\left. y = a - \frac{1}{a} - b + \frac{1}{b} \right\}$:

$vj := simplify(numer(\mathrm{expand}(\mathrm{subs}(\mathrm{parametrization, vj}))))$;

$$vj := 3\,a^4\,b^2 + (6\,b^3 - 6\,b)\,a^3 + (3\,b^4 - 28\,b^2 + 3)\,a^2 + (-6\,b^3 + 6\,b)\,a + 3\,b^2 \quad \textbf{(13)}$$

#We will check that vj=0 cuts it vertically the retangle [v , 3) $\times [1, v)$

#Apply the Möbius transformation on the rectangle $[v, 3) \times [1, v)$ *constructing* $\hat{vj}$

$v := \frac{1 + \mathrm{sqrt}(5)}{2}$:

$moebius := \left\{ a = \frac{\mathrm{v} + 3 \cdot \mathrm{c}}{1 + c}, \right.$
$\left. b = \frac{1 + \mathrm{v} \cdot d}{1 + d} \right\}$:

$\hat{\mathrm{vj}} := \mathrm{collect}(\mathrm{numer}(\mathrm{expand}(\mathrm{subs}(\mathrm{moebius, vj}))), c)$:

#Compute the degree of $\hat{vj}$ *in c*

$degree(\hat{vj}, c)$;

$$4 \quad \textbf{(14)}$$

#Define the coefficients ci = coeff$(\hat{vj}, c, i)$
$seq(evalf(assign(cat('c', i), coeff(\hat{vj}, c, i))), i = 0\,..\,4)$;

#Compute the all degree of ci in d
$seq(degree(eval(cat('c', i)), d), i = 0\,..\,4)$;

$$4, 4, 4, 4, 4 \quad \textbf{(15)}$$

#From here we will analyze the signal variation of all coefficients ci

$seq(sign(evalf(coeff(c4, d, i))), i = 4 .. 0, -1);$

$$1, 1, 1, 1, 1 \tag{16}$$

$seq(sign(evalf(coeff(c3, d, i))), i = 4 .. 0, -1);$
$fsolve(c3 = 0, d, 0 .. \text{infinity});$

$$1, 1, 1, 1, -1$$

$$0.1978860076 \tag{17}$$

$seq(sign(evalf(coeff(c2, d, i))), i = 4 .. 0, -1);$
$fsolve(c3 = 0, d, 0 .. \text{infinity});$

$$1, 1, -1, -1, -1$$

$$0.1978860076 \tag{18}$$

$seq(sign(evalf(coeff(c1, d, i))), i = 4 .. 0, -1);$
$fsolve(c1 = 0, d, 0 .. \text{infinity});$

$$1, -1, -1, -1, -1$$

$$800.9878211 \tag{19}$$

$seq(sign(evalf(coeff(c0, d, i))), i = 4 .. 0, -1);$

$$-1, -1, -1, -1, -1 \tag{20}$$

#This prove that vj=0 is a curve that cuts the retangle $[v, 3) \times [1, v)$ *vertically*

#Now we will check that r4=0 and vj=0 have at least one intersection in the retangle $[v, 3) \times [1, v)$ *where* $\gamma 4$ *are defined.*

#Let be a1 such that vj(a1,1)=0 and av such that $vj(av, v) = 0$.

#Check that a1< 2.75 < gamma4 <3

$sturm\left(sturmseq(eval(vj, b = 1), a), a, \frac{11}{4}, 3\right);$ #number of roots in (2.75 , 3)

$$0 \tag{21}$$

#Check that v< av <3

$sturm(sturmseq(eval(vj, b = v), a), a, \mathrm{v}, 3);$ #number of roots in (v , 3)

$$1 \tag{22}$$

#This prove the existence of an intersection. We will conclude the demonstration by showing the uniqueness.

#Define r4

$r4 := \left(a^{22} b^{12} + \left(8 b^{13} - 6 b^{11}\right) a^{21} + \left(30 b^{14} + 6 b^{12} + 15 b^{10}\right) a^{20} + \left(74 b^{15} + 162 b^{13} - 102 b^{11} - 18 b^{9}\right) a^{19} + \left(145 b^{16} + 600 b^{14} + 93 b^{12} + 198 b^{10} + 3 b^{8}\right) a^{18} + \left(252 b^{17} + 1380 b^{15} + 1884 b^{13} - 762 b^{11} - 96 b^{9} + 24 b^{7}\right) a^{17} + \left(145 b^{18} + 1092 b^{16} + 1908 b^{14} - 2920 b^{12}\right.$

$- 3162\,b^{10} - 1740\,b^{8} - 294\,b^{6}\big)\,a^{16} + \big(74\,b^{19} + 1380\,b^{17} + 4692\,b^{15} + 3340\,b^{13} - 8010\,b^{11}$
$- 2472\,b^{9} - 132\,b^{7} + 24\,b^{5}\big)\,a^{15} + \big(30\,b^{20} + 600\,b^{18} + 1908\,b^{16} - 3516\,b^{14} - 24921\,b^{12}$
$- 26352\,b^{10} - 11586\,b^{8} - 1740\,b^{6} + 3\,b^{4}\big)\,a^{14} + \big(8\,b^{21} + 162\,b^{19} + 1884\,b^{17} + 3340\,b^{15}$
$- 8268\,b^{13} - 34416\,b^{11} - 14520\,b^{9} - 2472\,b^{7} - 96\,b^{5} - 18\,b^{3}\big)\,a^{13} + b^{2}\,\big(b^{20} + 6\,b^{18} + 93\,b^{16}$
$- 2920\,b^{14} - 24921\,b^{12} - 68232\,b^{10} - 67416\,b^{8} - 26352\,b^{6} - 3162\,b^{4} + 198\,b^{2} + 15\big)\,a^{12}$
$- 6\,b\,\big(b^{20} + 17\,b^{18} + 127\,b^{16} + 1335\,b^{14} + 5736\,b^{12} + 11692\,b^{10} + 5736\,b^{8} + 1335\,b^{6} + 127\,b^{4}$
$+ 17\,b^{2} + 1\big)\,a^{11} + \big(15\,b^{20} + 198\,b^{18} - 3162\,b^{16} - 26352\,b^{14} - 67416\,b^{12} - 68232\,b^{10}$
$- 24921\,b^{8} - 2920\,b^{6} + 93\,b^{4} + 6\,b^{2} + 1\big)\,a^{10} + \big(-18\,b^{19} - 96\,b^{17} - 2472\,b^{15} - 14520\,b^{13}$
$- 34416\,b^{11} - 8268\,b^{9} + 3340\,b^{7} + 1884\,b^{5} + 162\,b^{3} + 8\,b\big)\,a^{9} + 3\,b^{2}\,\big(b^{16} - 580\,b^{14}$
$- 3862\,b^{12} - 8784\,b^{10} - 8307\,b^{8} - 1172\,b^{6} + 636\,b^{4} + 200\,b^{2} + 10\big)\,a^{8} + \big(24\,b^{17} - 132\,b^{15}$
$- 2472\,b^{13} - 8010\,b^{11} + 3340\,b^{9} + 4692\,b^{7} + 1380\,b^{5} + 74\,b^{3}\big)\,a^{7} + \big(-294\,b^{16} - 1740\,b^{14}$
$- 3162\,b^{12} - 2920\,b^{10} + 1908\,b^{8} + 1092\,b^{6} + 145\,b^{4}\big)\,a^{6} + \big(24\,b^{15} - 96\,b^{13} - 762\,b^{11}$
$+ 1884\,b^{9} + 1380\,b^{7} + 252\,b^{5}\big)\,a^{5} + \big(3\,b^{14} + 198\,b^{12} + 93\,b^{10} + 600\,b^{8} + 145\,b^{6}\big)\,a^{4} + \big($
$-18\,b^{13} - 102\,b^{11} + 162\,b^{9} + 74\,b^{7}\big)\,a^{3} + \big(15\,b^{12} + 6\,b^{10} + 30\,b^{8}\big)\,a^{2} + \big(-6\,b^{11} + 8\,b^{9}\big)\,a$
$+ b^{10}\big)$:

#Calculate the resultante of r4 and vj with respect to a

$Ra := resultant(r4, vj, a)$:

sturm(*sturmseq*(Ra, b) , b, 1, v); #number of roots in (1 , v)

$$1 \tag{23}$$

#Calculate the resultante of r4 and vj with respect to b

$Rb := resultant(r4, vj, b)$:

sturm(*sturmseq*(Rb, a) , a, v, 3); #number of roots in (v , 3)

$$1 \tag{24}$$

#The resultants of r4 and nj with respect to a and b admit a uniqueness root in the interval where a are defined for r4=0. This demonstrate the uniqueness.